Python and Machine Learning

Seminar 2: Basic probability theory and OLS

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Some definitions

In practice, causal inference is based on statistical models that range from the very simple to extremely advanced. Building such models requires some rudimentary knowledge of probability theory, so let's begin with some definitions.

- A **random process** is a process that can be repeated many times with different outcomes each time.
- The **sample space** is the set of all the possible outcomes of a random process.

Description	**Type**	**Possible outcomes**
12-sided die	Discrete	1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12
Coin	Discrete	Heads, Tails
Deck of cards	Discrete	2 $\diamondsuit$, 3 $\diamondsuit$, ..., King $\heartsuit$, Ace $\heartsuit$
Gas prices	Continuous	$P \geq 0$

Independent events

We define independent events two ways.

1. **Logical independence**. For instance, two events occur but there is no reason to believe that the two events affect each other.
2. **Statistical independence**. We'll illustrate the latter with an example from the idea of sampling with and without replacement.

Let's use a randomly shuffled deck of cards for an example. For a deck of 52 cards, what is the probability that the first card will be an ace?

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Let's use a randomly shuffled deck of cards for an example. For a deck of 52 cards, what is the probability that the first card will be an ace?

$$\Pr(\text{Ace}) = \frac{\text{Count of Aces}}{\text{Sample Space}} = \frac{4}{52} = 0.077$$

Independent events (cont'd)

Assume that the first card was an ace. Now we ask the question again. If we shuffle the deck, what is the probability the next card drawn is also an ace?

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It is no longer $\frac{4}{52}$ because we did not sample with replacement. We sampled *without* replacement. Thus the new probability is:

$$\Pr(\text{Ace} \mid \text{Card } 1 = \text{Ace}) = \frac{3}{51} = 0.059$$

- Two events, A and B, are independent if and only if:

$$\Pr(A|B) = \Pr(A)$$

Joint probability

But what if we want to know the probability of some event occurring that requires that multiple events first to occur?

In Texas Hold'em poker, each player is dealt two cards facedown. When you are holding two of a kind, you say you have two "in the pocket." So, what is the probability of being dealt pocket aces?

$$\frac{4}{52} \times \frac{3}{51} = 0.0045$$

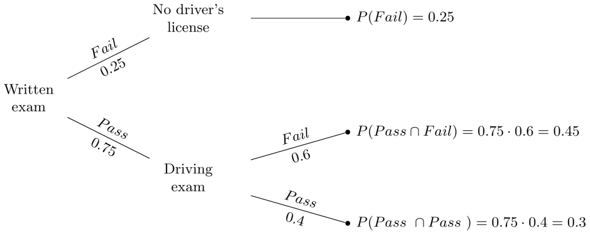
- For independent events, to calculate **joint probabilities**, we multiply the marginal probabilities:

$$\Pr(A, B) = \Pr(A)\Pr(B)$$

Conditional probability

Let A be some event. And let B be some other event. For two events, there are four possibilities:

1. A and B: Both A and B occur.
2. $\sim A$ and B: A does not occur, but B occurs.
3. A and $\sim B$: A occurs, but B does not occur.
4. $\sim A$ and $\sim B$: Neither A nor B occurs.



$$\Pr(A) = \sum_n \Pr(A \cap B_n)$$

Contingency tables

England's football coach has been on the razor's edge with the athletic director and regents after the last Euros. After several failures at the finals, his future with the team is in jeopardy. If the England national team doesn't win the Euros, he likely won't be rehired. But if they do, then it's likely that he will be rehired.

In this context, let A be England winning the Euros and B be the coach getting rehired. Consider $\Pr(A) = 0.6$, $\Pr(B) = 0.8$ and $\Pr(A, B) = 0.5$.

Event labels	**Coach is not rehired** $(\sim B)$	**Coach is rehired** (B)	**Total**
(A) Euros won	$\Pr(A, \sim B) = 0.1$	$\Pr(A, B) = 0.5$	$\Pr(A) = 0.6$
$(\sim A)$ Euros lost	$\Pr(\sim A, \sim B) = 0.1$	$\Pr(\sim A, B) = 0.3$	$\Pr(\sim A) = 0.4$
Total	$\Pr(\sim B) = 0.2$	$\Pr(B) = 0.8$	1

$$\Pr(B \mid A) = \frac{\Pr(A, B)}{\Pr(A)} = \frac{0.5}{0.6} = 0.83$$

Bayes's rule

So, we can use what we have done so far to write out a definition of joint probability. Let's start with a definition of conditional probability first. Given two events, A and B:

$$
\begin{aligned}
\Pr(A \mid B) &= \frac{\Pr(A, B)}{\Pr(B)} \\
\Pr(B \mid A) &= \frac{\Pr(B, A)}{\Pr(A)} \\
\Pr(A, B) &= \Pr(B, A) \\
\Pr(A) &= \Pr(A, \sim B) + \Pr(A, B) \\
\Pr(B) &= \Pr(A, B) + \Pr(\sim A, B)
\end{aligned}
$$

using the definitions of $\Pr(A, B)$ and $\Pr(B, A)$, we can also rearrange terms, make a substitution, and rewrite it as:

$$\Pr(A \mid B) \Pr(B) = \Pr(B \mid A) \Pr(A)$$

$$\Pr(A \mid B) = \frac{\Pr(B \mid A) \Pr(A)}{\Pr(B)}$$

Bayes's rule – example

Let's consider a medical test example. Given:

- $\Pr(D) = 0.01$
- $\Pr(\sim D) = 0.99$
- $\Pr(T|D) = 0.99$
- $\Pr(\sim T| \sim D) = 0.95$

Find the posterior probability $\Pr(D|T)$.

$$
\begin{aligned}
\Pr(D|T) &= \frac{\Pr(T|D)\Pr(D)}{\Pr(T)} \\
\Pr(T) &= \Pr(T|D)\Pr(D) + \Pr(T| \sim D)\Pr(\sim D) \\
&= (0.99 \times 0.01) + (0.05 \times 0.99) \\
&= 0.0099 + 0.0495 \\
&= 0.0594 \\
\Pr(D|T) &= \frac{0.99 \times 0.01}{0.0594} \\
&= \frac{0.0099}{0.0594} \\
&= 0.1667
\end{aligned}
$$

Thus, the probability of having the disease given a positive test result is approximately 0.167 or 16.67%.

Expected value

The expected value of a **random variable**, also sometimes called the population mean, is simply the weighted average of the possible values that the variable can take, with the **weights** being given by the probability of each value occurring in the population.

Suppose that the variable X can take on values $x_1, x_2, \ldots, x_k$, each with probability $f(x_1), f(x_2), \ldots, f(x_k)$, respectively. Then we define the expected value of X as:

$$\begin{aligned} E(X) &= x_1 f(x_1) + x_2 f(x_2) + \cdots + x_k f(x_k) \\ &= \sum_{j=1}^{k} x_j f(x_j) \end{aligned}$$

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3. If we have numerous constants $a_1, \ldots, a_n$, and many random variables $X_1, \ldots, X_n$ then

$$E(a_1X_1 + \cdots + a_nX_n) = a_1E(X_1) + \cdots + a_nE(X_n)$$

which can also be expressed as:

$$E\left(\sum_{i=1}^{n} a_iX_i\right) = \sum_{i=1} a_iE(X_i)$$

Variance of a random variable

The expectation operator, $E(.)$, is a **population concept**. It refers to the whole group of interest, not just to the sample available to us. Its meaning is somewhat similar to that of the average of a random variable in the population.

Some additional properties for the expectation operator can be explained assuming two random variables, W and H.

$$E(aW + b) = aE(W) + b \text{ for any constants } a,\, b$$

$$E(W + H) = E(W) + E(H)$$

$$E\Big(W - E(W)\Big) = 0$$

Consider the **variance** of a random variable, W:

$$\text{Var}(W) = \sigma^2 = E\Big[\big(W - E(W)\big)^2\Big] \text{ in the population}$$

$$\text{Var}(W) = E(W^2) - E(W)^2$$

Variance – population vs. sample

1. **Population variance**. To calculate the population variance, we use the formula:

$$\sigma^2 = \frac{1}{N}\sum_{i=1}^{N}(X_i - \mu)^2$$

where N is the size of the population consisting of $X_1, X_2, \ldots X_N$, and μ is the population mean.

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Make sure you know when to make this distinction. To use the **population variance** you need all of the data available whereas to use the **sample variance** you only need a proportion of it. For example, if we take ten words at random from a Wikipedia page to calculate the variance of their length, a sample variance would be needed. To find the population variance, the length of every word on the Wikipedia page would be needed.

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If the two variables are **independent**, then $E(XY) = E(X)E(Y)$ and $\mathrm{Var}(X + Y)$ is equal to the sum of $\mathrm{Var}(X) + \mathrm{Var}(Y)$

Exercise

Consider a random variable X representing the scores of students in an exam, where X follows a normal distribution with mean, $\mu = 70$ and variance, $\sigma^2 = 25$.

- **Expected value:**
  1. Compute the expected value $E[X]$.
  2. If each student's score is increased by 5 points as a bonus, what will be the new expected value $E[X+5]$?
- **Variance:**
  1. Compute the variance $\text{Var}(X)$.
  2. If each student's score is increased by 5 points as a bonus, what will be the new variance $\text{Var}(X+5)$?
  3. If each student's score is multiplied by 2, what will be the new variance $\text{Var}(2X)$?
  4. Given a sample of $n = 10$ students with the following scores: 65, 70, 75, 60, 80, 85, 90, 55, 70, 75. Compute the sample mean $\bar{x}$.
  5. Compute the sample variance s^2.

Exercise – solutions

Expected value:

1. $E[X] = \mu = 70$
2. $E[X+5] = E[X] + 5 = 70 + 5 = 75$

Variance:

1. $\mathrm{Var}(X) = \sigma^2 = 25$
2. $\mathrm{Var}(X+5) = \mathrm{Var}(X) = 25$
3. $\mathrm{Var}(2X) = 4 \cdot \mathrm{Var}(X) = 4 \cdot 25 = 100$
4. Sample mean $\bar{x}$:

$$\bar{x} = \frac{1}{10}\sum_{i=1}^{10} x_i = \frac{65+70+75+60+80+85+90+55+70+75}{10} = \frac{725}{10} = 72.5$$

5. Sample variance s^2:

$$s^2 = \frac{1}{9}\left((65-72.5)^2 + (70-72.5)^2 + (75-72.5)^2 + (60-72.5)^2 + (80-72.5)^2 + (85-72.5)^2 + (90-72.5)^2 + (55-72.5)^2 + (70-72.5)^2 + (75-72.5)^2\right)$$

$$s^2 = \frac{1}{9}(56.25 + 6.25 + 6.25 + 156.25 + 56.25 + 156.25 + 306.25 + 306.25 + 6.25 + 6.25) = 118.06$$

Covariance

$$\text{Var}(X+Y) = \text{Var}(X) + \text{Var}(Y) + 2\Big(E(XY) - E(X)E(Y)\Big)$$

Recall the equation above. The last part of the equation is called the **covariance**. The covariance measures the amount of linear dependence between two random variables. We represent it with the $\text{Cov}(X, Y)$ operator.

The expression $\text{Cov}(X, Y) > 0$ indicates that two variables move in the same direction, whereas $\text{Cov}(X, Y) < 0$ indicates that they move in opposite directions. We can rewrite the equation above as:

$$\text{Var}(X+Y) = \text{Var}(X) + \text{Var}(Y) + 2\text{Cov}(X, Y)$$

- While it's tempting to say that a zero covariance means that two random variables are unrelated, that is incorrect. They could have a **nonlinear relationship**.
- If X and Y are independent, then $\text{Cov}(X, Y) = 0$ in the population.

Class size and performance

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- Class size reduction has been at the heart of policy debates for decades.
- We will be using data from a famous paper by Joshua Angrist and Victor Lavy (1999).
- Consists of test scores and class/school characteristics for fifth graders (10-11 years old) in Jewish public elementary schools in Israel in 1991.
- National tests measured mathematics and (Hebrew) reading skills. The raw scores were scaled from 1-100.

Task 1: Getting to Know the Data

1. Load the data from Moodle as `grades`. *Hint: Use the* ***`read_stata`*** *function from the* ***`pandas`*** *library to import the file, which has a format .dta. (FYI: .dta is the extension for data files used in Stata).*
2. Describe the data:
   - What is the unit of observation, i.e., what does each row correspond to?
   - How many observations are there?
   - View the dataset. What variables do we have? What do the variables `avgmath` and `avgverb` correspond to?
   - Use the `describe()` method from `pandas` to obtain common summary statistics for the variables `classize`, `avgmath`, and `avgverb`. *(Hint: use* ***`pandas`*** *to select the variables and then call* ***`grades[['classize', 'avgmath', 'avgverb']].describe()`****).*
3. Do you have any priors about the actual (linear) relationship between class size and student achievement? What would you do to get a first insight?
4. Compute the correlation between class size and math and verbal scores using `pandas.corr()` method. Is the relationship positive/negative, strong/weak?

Regressions - population model

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This model is assumed to hold in the population. The equation defines a **linear regression model**. For models concerned with capturing causal effects, the terms on the left side are usually thought of as the effect, and the terms on the right side are thought of as the causes.

The error term - u

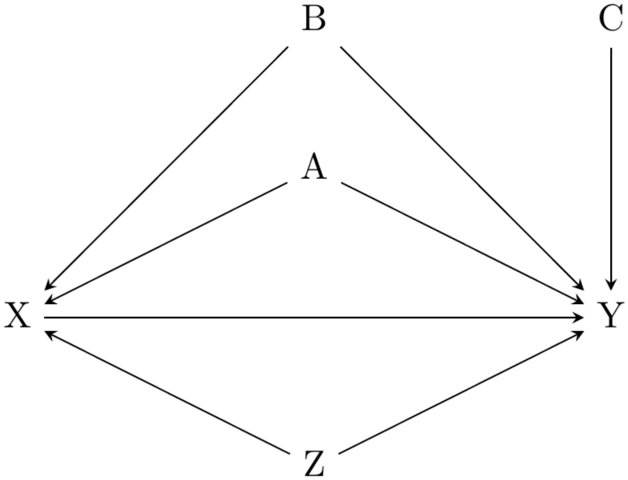

Figure 1: Depending on the model used, A, B, C and Z might be in the error term.

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- This equation also explicitly models the functional form by assuming that Y is **linearly dependent** on X.
- We call the β_0 coefficient the intercept parameter, and we call the β_1 coefficient the slope parameter.
  - These describe a population, and our goal in empirical work is to **estimate** their values.
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  - We never directly observe these parameters, because they are not data.
  - We estimate the parameters using data and assumptions.
- In this simple regression framework, all unobserved variables that determine y are subsumed by the error term u.
- The first assumption we make is that $E(u) = 0$, **normalization of errors**.

Mean independence

Another main assumption we make is that the error mean is independent:

$$E(u \mid x) = E(u) \text{ for all values } x$$

where $E(u \mid x)$ means "the expected value of u given x". If the equation holds, then we say that u is mean independent of x.

Let's say we are estimating the effect of years of schooling on wages, and u is unobserved ability. Mean independence requires that

$$E(\text{ability} \mid x = 8) = E(\text{ability} \mid x = 12) = E(\text{ability} \mid x = 16)$$

so that the average ability is the same in the different portions of the population with different years of schooling. Because people choose how much schooling to invest in based on their own unobserved skills and attributes, the mean independence assumption is likely violated–at least in our example.

Let's say we're willing to make this assumption anyways. Combining it with the **normalization of errors** assumption, we get:

$$E(u \mid x) = 0, \text{ for all values } x$$

This equation is called the **zero conditional mean** assumption. Because the conditional expected value is a linear operator, this implies that:

$$E(y \mid x) = \beta_0 + \beta_1 x$$

Ordinary Least Squares

Let $\{(x_i, \text{ and } y_i) : i = 1, 2, \ldots, n\}$ be random samples of size n from the population. Plug any observation into the population equation:

$$y_i = \beta_0 + \beta_1 x_i + u_i$$

where i indicates a particular observation. We observe y_i and x_i but not u_i. We just know that u_i is there. We then use the two population restrictions that we discussed earlier:

$$E(u) = 0$$
$$E(u \mid x) = 0$$

and replacing u with $y - \beta_0 - \beta_1 x$:

$$E(y - \beta_0 - \beta_1 x) = 0$$
$$\Big(x[y - \beta_0 - \beta_1 x]\Big) = 0$$

These are the two conditions in the population that effectively determine β_0 and β_1. Note that the notation here is population concepts. We don't have access to populations, though we do have their sample counterparts:

$$\frac{1}{n} \sum_{i=1}^{n} \left(y_i - \widehat{\beta_0} - \widehat{\beta_1} x_i \right) = 0$$

$$\frac{1}{n} \sum_{i=1}^{n} \left(x_i \left[y_i - \widehat{\beta_0} - \widehat{\beta_1} x_i \right] \right) = 0$$

Ordinary Least Squares (cont'd)

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After multiple other theoretical steps that we are not concerned about in this class, we reach the following equation using estimates $\widehat{\beta}_0, \widehat{\beta}_1$ that defines a fitted value for each i:

$$\widehat{y_i} = \widehat{\beta}_0 + \widehat{\beta}_1 x_i$$

Recall that $i = \{1, \ldots, n\}$, so we have n of these equations. This is the value we predict for y_i given that $x = x_i$. But there is prediction error because $y \neq y_i$. We call that mistake the residual, and here use the $\widehat{u_i}$ for it. The residual equals:

$$\widehat{u_i} = y_i - \widehat{y_i}$$
$$\widehat{u_i} = y_i - \widehat{\beta_0} - \widehat{\beta_1} x_i$$

- While both the **residual** and the **error term** are represented with a u, it is important that you know the differences.
- The residual is the prediction error based on our fitted $\widehat{y_i}$ and the actual y_i.
- But u without the hat is the error term, and it is by definition unobserved by the researcher.

Ordinary Least Squares (cont'd)

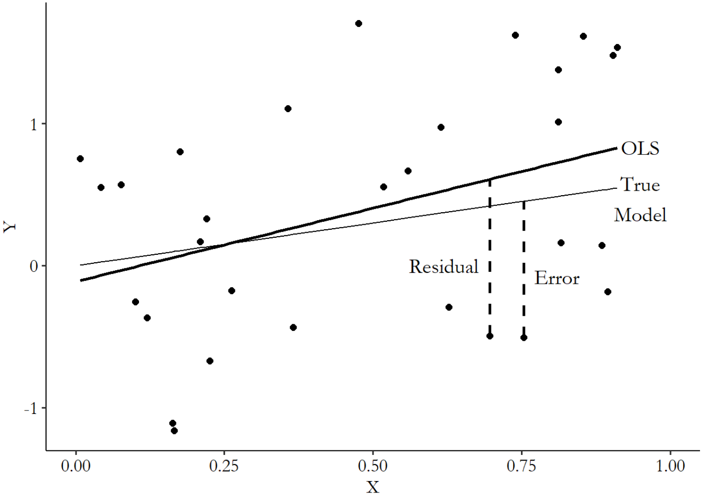


Figure 2: The difference between the residual and the error.

Ordinary Least Squares (cont'd)

Suppose we measure the size of the mistake, for each i, by **squaring** it. Squaring it will eliminate all negative values of the mistake so that everything is a positive value. This becomes useful when summing the mistakes if we don't want positive and negative values to cancel one another out. So let's do that: square the mistake and add them all up to get:

$$\begin{aligned}\sum_{i=1}^{n} \widehat{u_i}^2 &= \sum_{i=1}^{n} (y_i - \widehat{y_i})^2 \\ &= \sum_{i=1}^{n} \left(y_i - \widehat{\beta_0} - \widehat{\beta_1} x_i\right)^2\end{aligned}$$

This equation is called the **sum of squared residuals** because the residual is $\widehat{u_i} = y_i - \widehat{y}$. But the residual is based on estimates of the slope and the intercept. We can imagine any number of estimates of those values. Our goal is to minimize the sum of squared residuals by choosing $\widehat{\beta_0}$ and $\widehat{\beta_1}$.

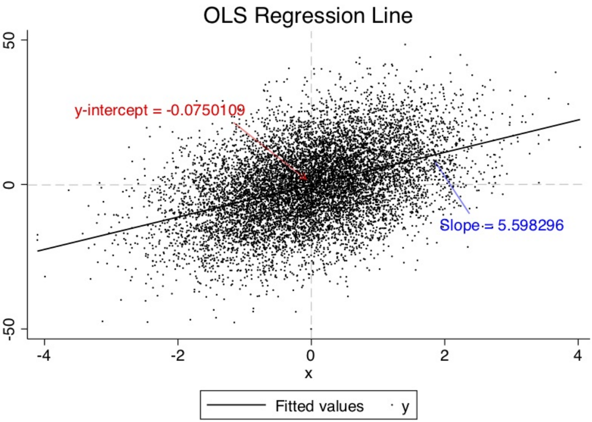
Once we have the numbers $\widehat{\beta_0}$ and $\widehat{\beta_1}$ for a given data set, we write the OLS regression line:

$$\widehat{y} = \widehat{\beta_0} + \widehat{\beta_1} x$$

In a **single independent variable** case:

$$\widehat{\beta_0} = \bar{y} - \widehat{\beta_1}\bar{x} \quad \text{and} \quad \widehat{\beta_1} = \frac{cov(x,y)}{var(x)}$$

Ordinary Least Squares - simulation



Let's try to recreate this simulation in Python.

Task 2: OLS Regression

Run the following code to aggregate the data at the class size level:

```
grades_avg_cs = grades.groupby('classize').agg(
    avgmath_cs=('avgmath', 'mean'),
    avgverb_cs=('avgverb', 'mean')
).reset_index()
```

1. Regress average verbal score (dependent variable) on class size (independent variable). Interpret the coefficients. scikit-learn OLS
2. Compute the OLS coefficients $\widehat{\beta_0}$ and $\widehat{\beta_1}$ of the previous regression using the formulas on slide 25.
3. What is the predicted average verbal score when class size is equal to 0? *Does that even make sense?*
4. What is the predicted average verbal score when the class size is equal to 30 students?

Goodness of fit

- The R^2 measures how well the **model fits the data**.

$$R^2 = \frac{\text{variance explained}}{\text{total variance}} = \frac{\text{SSE}}{\text{SST}} = 1 - \frac{\text{SSR}}{\text{SST}} \in [0, 1]$$

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- R^2 close to 1 indicates a **very high explanatory power** of the model.
- R^2 close to 0 indicates a **very low explanatory power** of the model.

Goodness of fit

- The R^2 measures how well the **model fits the data**.

$$R^2 = \frac{\text{variance explained}}{\text{total variance}} = \frac{\text{SSE}}{\text{SST}} = 1 - \frac{\text{SSR}}{\text{SST}} \in [0, 1]$$

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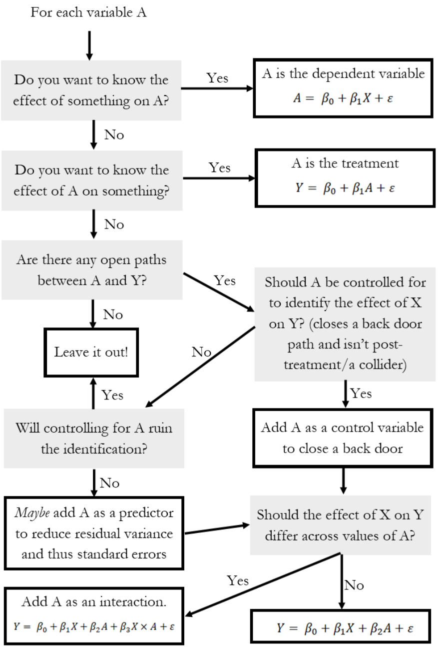
- Low R^2 does **NOT** mean it's a useless model. Remember that econometrics is interested in causal mechanisms, not prediction!
- The R^2 is **NOT** an indicator of whether a relationship is causal!

Task 3: R^2 and Goodness of Fit

1. Regress `avgmath_cs` on `classize` using the `scikitlearn` library. Assign the regression results to an object `math_reg`.
2. Use the `summary()` method on `math_reg`. What is the R^2 for this regression? How can you interpret it?
3. Compute the squared correlation between `classize` and `avgmath_cs`. What does this tell you about the relationship between R^2 and the correlation in a regression with only one regressor?
4. Repeat steps 1 and 2 for `avgverb_cs`. For which exam does the variance in class size explain more of the variance in students' scores?


For each variable A
Do you want to know the effect of something on A?
Yes
A is the dependent variable
$A = \beta_0 + \beta_1 X + \varepsilon$
No
Do you want to know the effect of A on something?
Yes
A is the treatment
$Y = \beta_0 + \beta_1 A + \varepsilon$
No
Are there any open paths between A and Y?
Yes
Should A be controlled for to identify the effect of X on Y? (closes a back door path and isn't post-treatment/a collider)
No
Leave it out!
No
Yes
Yes
Will controlling for A ruin the identification?
Add A as a control variable to close a back door
No
Maybe add A as a predictor to reduce residual variance and thus standard errors
Should the effect of X on Y differ across values of A?
Yes
No
Add A as an interaction.
$Y = \beta_0 + \beta_1 X + \beta_2 A + \beta_3 X \times A + \varepsilon$
$Y = \beta_0 + \beta_1 X + \beta_2 A + \varepsilon$

Example - Education and wages

Let's consider the following example concerning wage data collected in the 1976 Current Population Survey in the USA. We want to investigate the relationship between average hourly earnings, and years of education.

1. Plot the distribution of wages and years of education vs. wages.
   - We see that wages are very concentrated at around 5 USD per hour, with fewer and fewer observations at higher rates.
   - The hourly wage seems to increase with higher education levels.

2. Run the following regression:

$$wage_i = b_0 + b_1 educ_i + e_i$$

We are interested in the relationship between two variables:

- an **outcome variable** (also called **dependent variable**): *wage (y)*
- an **explanatory variable** (also called **independent variable**): *education (x)*

Example - Interpretation

Table 1: OLS results.

	Dependent variable:
	wage
educ	0.541^{***}
	(0.053)
Constant	-0.905
	(0.685)
Observations	526
R^2	0.165
Adjusted R^2	0.163
Residual Std. Error	3.378 (df = 524)
F Statistic	103.363^{***} (df = 1; 524)
Note:	*p<0.1; **p<0.05; ***p<0.01

The main interpretation of this table can be read off the column labelled Estimate (in R), reporting estimated coefficients b_0, b_1:

1. With zero year of education, the hourly wage is about -0.9$ per hour.
2. Each additional year of education increase hourly wage by 54 cents.
3. For example, for 15 years of education, we predict roughly $-0.9 + 0.541 * 15 = 7.215\$/h$.

Example - The log-transform

The natural logarithm is a particularly important transformation that we often encounter in economics. Why would we transform a variable with the log function to start with?

1. Several important economic variables (like wages, city size, firm size, etc) are approximately log-normally distributed. By transforming them with the log, we obtain an approximately normally distributed variable, which has desirable properties for our regression.
2. Applying the log reduces the impact of outliers.
3. The transformation allows for a convenient interpretation in terms of percentage changes of the outcome variable.

Modifying our original equation to apply log-transformation:

$$\log(wage_i) + b_0 + b_1 educ_i + e_i$$

Example - The log-transform

Table 2: OLS results with both the standard and the log-transformed variable.

	Dependent variable:	
	wage	log(wage)
	(1)	(2)
educ	0.541^{***}	0.083^{***}
	(0.053)	(0.008)
Constant	-0.905	0.584^{***}
	(0.685)	(0.097)
Observations	526	526
R^2	0.165	0.186
Adjusted R^2	0.163	0.184
Residual Std. Error (df = 524)	3.378	0.480
F Statistic (df = 1; 524)	103.363^{***}	119.582^{***}
Note:	$^{*}p<0.1$; $^{**}p<0.05$; $^{***}p<0.01$	

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- Still, the model is probably biased because it's missing other relevant information that would explain wages. This bias is called the **Ommitted Variables Bias (OVB)**.

Log-transformation

Table 3: Common Regression Specifications

Specification	**Outcome Var**	**Regressor**	**Interpretation of b_1**
Level-level	y	x	$\Delta y = b_1 \Delta x$
Level-log	y	$\log(x)$	$\Delta y = \frac{b_1}{100} \Delta x$
Log-level	$\log(y)$	x	$\% \Delta y = (100 b_1) \Delta x$
Log-Log	$\log(y)$	$\log(x)$	$\% \Delta y = \% \Delta b_1 \times x$

A tale of two colleges

Students who attended private four-year colleges in the US paid about $29,000 in tuition and fees during the 2012-2013 school year, while public university students in their home states paid less than $9,000.

While private schools may offer smaller class sizes, newer athletic facilities, more distinguished faculty, and possibly smarter peers, the critical question remains: Does this investment yield higher earnings in the long term?

- How much would a 40-year-old graduate of Harvard have earned if they had instead attended U-Mass?
- There is a selection bias here → difficult to compare earnings.
- To estimate the earnings effect of attending an elite school, we need to control for these pre-existing differences.

A tale of two colleges: comparing students

		Private			Public			
Applicant group	Student	Ivy	Leafy	Smart	All State	Tall State	Altered State	1996 earnings
A	1		Reject	Admit		Admit		110,000
	2		Reject	Admit		Admit		100,000
	3		Reject	Admit		Admit		110,000
B	4	Admit			Admit		Admit	60,000
	5	Admit			Admit		Admit	30,000
C	6		Admit					115,000
	7		Admit					75,000
D	8	Reject			Admit	Admit		90,000
	9	Reject			Admit	Admit		60,000

(a) Enrollment decisions are highlighted in gray.

Within each group, students are likely to have similar career ambitions, while they were also judged to be of similar ability by admissions staff at the schools to which they applied. Within-group comparisons should therefore be considerably more apples-to-apples than uncontrolled comparisons involving all students.

A tale of two colleges: comparing students (cont'd)

- **Group A**: private vs. public choice
  - Three students applied to two private schools (Leafy, Smart) and one public school (Tall State).
  - All admitted to Smart and Tall State, but rejected by Leafy.
  - Two attended Smart (private), one chose Tall State (public) to save money.
  - **Private school differential**: -$5,000, indicating no advantage for private school in this group.

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- **Group B**: private advantage
  - Two students applied to one private (Ivy) and two public schools (All State, Altered State).
  - Both admitted to all schools. One attended Ivy (private), the other attended Altered State (public).
  - **Private school differential**: +\$30,000, suggesting a significant advantage for private school attendees.

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- **Group C & D**: uninformative
  - Group C: Both students applied to and attended the same private school (Leafy), providing no comparative data on public vs. private outcomes.
  - Group D: Both attended public schools (All State, Tall State), also uninformative for private school comparison.

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The average effect, combining Group A's -\$5,000 and Group B's +\$30,000, is \$12,500. This suggests that, on average, attending a private school leads to higher earnings, controlling for applicants' choices and abilities.

A tale of two colleges: simple regression

The key ingredients in the regression recipe are:

- the **dependent variable**, in this case, student i's earnings later in life, also called the **outcome variable** (denoted by Y_i);
- the **treatment variable**, in this case, a dummy variable that indicates students who attended a private college or university (denoted by P_i);
- a set of **control variables**, in this case, variables that identify sets of schools to which students applied and were admitted.

The regression equation can be written as:

$$Y_i = \alpha + \beta P_i + \gamma A_i + e_i$$

Running the regression on data for the five students in groups A and B generates the following estimates:

$$\alpha = 40,000$$
$$\beta = 10,000$$
$$\gamma = 60,000$$

A tale of two colleges: a more sophisticated regression

Let's look at another regression model, that might eliminate more of the selection bias.

$$\ln Y_i = \alpha + \beta P_i + \sum_{j=1}^{150} \gamma_j GROUP_{ji} + \delta_1 SAT_i + \delta_2 \ln PI_i + \theta \boldsymbol{X} + e_i.$$

- P_i – attendence at private school
- $GROUP_{ji}$ – if student i is in selectivity group j (groups based on selection choices and acceptance)
- SAT_i – SAT score of student i divided by 100
- $\ln PI_i$ – logarithm of parental income of student i
- $\boldsymbol{X}$ – other controls like gender, ethnicity etc.

A tale of two colleges: results table

	No selection controls			Selection controls		
	(1)	(2)	(3)	(4)	(5)	(6)
Private school	.135 (.055)	.095 (.052)	.086 (.034)	.007 (.038)	.003 (.039)	.013 (.025)
Own SAT score ÷ 100		.048 (.009)	.016 (.007)		.033 (.007)	.001 (.007)
Log parental income			.219 (.022)			.190 (.023)
Female			−.403 (.018)			−.395 (.021)
Black			.005 (.041)			−.040 (.042)
Hispanic			.062 (.072)			.032 (.070)
Asian			.170 (.074)			.145 (.068)
Other/missing race			−.074 (.157)			−.079 (.156)
High school top 10%			.095 (.027)			.082 (.028)
High school rank missing			.019 (.033)			.015 (.037)
Athlete			.123 (.025)			.115 (.027)
Selectivity-group dummies	No	No	No	Yes	Yes	Yes

A tale of two colleges: results

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We can explore this hypothesis by replacing the private school dummy in the model with a measure of peer quality. Specifically, we replace P_i in the equation with the average SAT score of classmates at the school attended.

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Columns (1)–(3) of the next table show that students who attended more selective schools do markedly better in the labor market, with an estimated college selectivity effect on the order of 8% higher earnings for every 100 points of average selectivity increase. Yet, this effect too appears to be an artifact of selection bias due to the greater ambition and ability of those who attend selective schools.

	No selection controls			Selection controls		
	(1)	(2)	(3)	(4)	(5)	(6)
School average SAT score ÷ 100	.109	.071	.076	−.021	−.031	.000
	(.026)	(.025)	(.016)	(.026)	(.026)	(.018)
Own SAT score ÷ 100		.049	.018		.037	.009
		(.007)	(.006)		(.006)	(.006)
Log parental income			.187			.161
			(.024)			(.025)
Female			−.403			−.396
			(.015)			(.014)
Black			−.023			−.034
			(.035)			(.035)
Hispanic			.015			.006
			(.052)			(.053)
Asian			.173			.155
			(.036)			(.037)
Other/missing race			−.188			−.193
			(.119)			(.116)
High school top 10%			.061			.063
			(.018)			(.019)
High school rank missing			.001			−.009
			(.024)			(.022)
Athlete			.102			.094
			(.025)			(.024)
Average SAT score of schools applied to ÷ 100				.138	.116	.089
				(.017)	(.015)	(.013)
Sent two applications				.082	.075	.063
				(.015)	(.014)	(.011)
Sent three applications				.107	.096	.074